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Decentralised Collaboration and AI-Mediated Practices in a Member-Governed Makerspace

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Abstract: This study asks how makers describe the collaborative practices associated with their member-governed makerspace's long-term survival, and how AI tools support, complicate or threaten those practices. This qualitative single-site case study draws on interview data collected from a UK member-governed makerspace. We performed abductive reflexive thematic analysis informed by Ostrom's (1990) commons governance framework, self-determination theory (SDT), and sociomateriality. The findings show that formal rules and trustee structures were necessary but insufficient, as coordination also depended on reputational authority, or, as a participant dubbed it, "*mana*," earned through visible contribution and accountable judgement. Participation was sustained through *mutual witnessing*, whereby members made individual projects visible and socially meaningful. Participants accepted AI for bounded tasks including search and prototyping but resisted uses that displaced traceable human judgement or collective deliberation. The findings position visible, accountable participation as a link between community governance, motivation, and negotiated boundaries around AI use.

Introduction

Makerspaces are settings in which people learn by designing, making, sharing, and revising artefacts through interactions with others (Sheridan et al., 2014). Makerspaces can be considered both *third places*—settings distinct from home, formal education, and work, where members return not because they are required to do so, but because the space continues to give them reasons to come back (Lister, 2026)—and *third spaces*—settings where collaborative interaction supports rich learning and the negotiation of sociocritical competencies (Gutiérrez, 2008; Kajamaa & Kumpulainen, 2020). While some makerspaces are embedded within established organisations, such as schools, museums, and libraries, others are independent, community-based and governed by members without pay or formal managerial hierarchies. The continuation of member-governed makerspaces depends on members contributing time, expertise, and care to the space (Jenkins et al., 2020; Toombs et al., 2015). This makes member-governed makerspaces demanding settings for collective action and collaborative learning, because participation cannot be compelled and the space must coordinate access to scarce resources such as tools, rooms, and members' time (Ostrom, 1990).

Of course, the absence of a designated leader does not necessarily mean the absence of leadership. Studies of leaderless or self-organising groups show that communities often develop distributed forms of leadership through everyday negotiation, consensus-building, and the informal authority of long-standing members (Choi & Schnurr, 2014; Sutherland et al., 2014). Within such member-governed commons, members coordinate access not only to tools and space but also to the expertise and shared knowledge that make participation possible (Hess & Ostrom, 2007; Ostrom, 1990). In this context, the maintenance labour sustaining these makerspace communities, such as welcoming newcomers, mediating conflict, and repairing shared infrastructure, is often concentrated invisibly among a few members (Toombs et al., 2015). From a situated learning perspective, this labour also gives newcomers legitimate access to a community of practice (Lave & Wenger, 1991; Wenger, 1998).

The rise of generative AI introduces further questions about how labour and responsibilities are distributed. AI tools may support prototyping, information retrieval, and coordination, while making the origins of information and responsibility for its use less clear. That is, who verifies AI-generated

information before it is shared, and who remains responsible for its use? Existing work on AI teammates is largely grounded in formal settings where a manager, protocol, or organisational structure can assign roles (Dennis et al., 2023; Siemon, 2022). Less is known about how AI is admitted, limited, or refused in informal and voluntary communities whose authority is negotiated collaboratively through everyday practice.

This study examines a UK makerspace that has maintained a substantial and active membership over many years without a conventional managerial hierarchy. It is guided by two main research questions:

1. How do makerspace participants describe collaborative practices associated with the makerspace's continued operation?
2. How do makerspace participants describe AI as supporting, complicating, or threatening those practices?

This study treats sustainability of the makerspace community as an emic and analytical concern. It examines conditions participants associate with continued community life as a collective and collaborative learning environment and explores how a local learning community negotiates the terms on which emerging technologies enter its practices.

Theoretical framework

To examine how an informal makerspace sustains participation and collaborative learning while negotiating the role of generative AI, this study brings together three complementary theoretical lenses: commons governance, self-determination theory (SDT), and sociomateriality. These lenses are used together to analyse how shared resources are governed, how members remain motivated to participate, and how AI becomes part of everyday practice.

Ostrom's (1990) commons governance framework, developed from studies of natural-resource commons, holds that the durability of a shared resource depends not on the absence of rules but on locally legitimate arrangements. These arrangements often include clearly defined boundaries, rules matched to local conditions, arrangements that allow participants to modify the rules, monitoring, graduated sanctions, accessible conflict resolution, and recognition of users' rights to organise. The framework has also been extended to knowledge commons, where access, contribution, and stewardship require collective coordination (Hess & Ostrom, 2007). In a makerspace, tools and rooms are rival resources, whereas expertise can be shared but still depends on members' limited time and attention. Commons governance therefore provides us with a vocabulary for analysing coordination without assuming either a managerial hierarchy or an organisational vacuum.

SDT identifies autonomy, competence, and relatedness as basic psychological needs that support functioning, growth, and social development, which helps to explain sustained voluntary participation (Ryan & Deci, 2000, 2017). In makerspaces, these needs involve choosing projects and influencing decisions, developing and demonstrating capability, and being recognised as a valued community member. Han et al. (2017) associated support for these needs with intrinsic motivation and continuance intention in makerspaces. Although their model treats these needs as additive predictors of intrinsic motivation and continuance intention, SDT does not require them to operate independently (Sheldon & Niemiec, 2006). In this study, SDT is used as a sensitising framework for analysing how these needs work together within voluntary participation and how practices involving AI may support or constrain autonomy, competence, and relatedness within makerspaces.

A sociomaterial perspective views social and material dimensions as mutually constitutive, rejecting the idea that technologies' effects arise from their features alone (Leonardi, 2013; Orlikowski, 2007). Prior CSCL research has found sociomateriality to be a useful lens for understanding collaborative knowledge construction, co-invention, and self-organisation within digitally mediated maker projects (e.g., Davies et al., 2024; Riikonen et al., 2020). Following from this work, we attend to how the properties of people and AI technologies enable or constrain practice. This includes examining how communities assign AI technologies roles, embed them in routines, and govern their use.

Together, the three lenses connect the governance, motivation, and technological organisation of voluntary participation within community-based makerspaces to help us describe the collaborative practices that sustain community activity and learning, and how AI functions as support, complication, or threat to those practices, thus addressing our research questions.

Methods

The study adopted an interpretivist, qualitative single-site case study design. The case is that of a community makerspace in the UK that operates through member participation and distributed governance without a conventional managerial hierarchy. Nine participants were recruited purposively to capture variation in involvement. The participants included eight current members ranging from elected trustees and long-standing members to recent joiners, and one visitor who completed a tour but did not join. Table 1 shows the range of tenure and roles. Semi-structured interviews ranged from approximately 34 to 107 minutes and covered coordination, trust, conflict, contribution, knowledge sharing, motivation, and AI use. The non-member account was retained as a limited contrast case rather than as evidence of why people disengage.

Table 1

Participant Overview

Participant	Role/Identity	Years in the makerspace
P1	Long-standing member	14 years
P2	Trustee	Over 10 years
P3	Member	7 years
P4	Member	1.5 years
P5	Non-member	Tour only
P6	Member	2 years
P7	Trustee & Active member	Approx. 4 years
P8	Member (recent joiner)	Under 1 year
P9	Member & Team member	Approx. 2 years

Recruitment used posters displayed inside the makerspace and an online call posted in the members' Discord channel, allowing both frequent and less frequent visitors to respond. The first author was also a member of the makerspace. This insider position supported access and interpretation. Reflexive journaling and an audit trail documented recruitment, coding, theme development, and the possible influence of familiarity with the setting. Ethical approval was obtained before data collection; participants and the makerspace were pseudonymised, and written informed consent was obtained.

Interview transcripts were analysed in MAXQDA using reflexive thematic analysis (Braun & Clarke, 2022), which treats themes as actively constructed through the researcher's interpretive engagement with the data rather than as discovered within it. Analysis was abductive, moving between data and theory, and predominantly inductive. Commons governance, SDT, and sociomateriality acted as sensitising concepts rather than as a fixed coding scheme. The corpus contained 959 coded excerpts across 85 original codes, organised into 15 initial themes which were then refined as sub-themes within three overarching theory-informed themes and one background category. Comparing related sub-themes across the dataset produced these three themes. Governance sub-themes concerning collective choice, monitoring, and continuity informed the theme of reputational authority beneath formal governance. Autonomy, competence, and relatedness were read alongside accounts of individual projects and entry to inform the theme of sustained participation through mutual witnessing. AI-use, attitude, and sociomaterial sub-themes were compared with accounts of verification and responsibility to develop the theme of selective absorption of AI. Frequencies were used only as a descriptive audit of coverage; they were not treated as measures of thematic importance, saturation, or prevalence. The

study captures reported experiences and interpretations within one site and does not establish causal effects of AI or statistical generalisability.

Findings

In this section, we present the three themes derived from the analysis described above. These findings describe patterns and tensions within participants' accounts related to 1) reputational authority beneath formal governance, 2) sustained participation through *mutual witnessing*, and 3) selective absorption of AI. The first two themes address the first research question, and the third theme addresses the second research question.

Reputational authority beneath formal governance

This theme examines how governance shaped access, accountability, and sustained participation over time. Participants described membership rules, an elected trustee board, tool teams, monitoring, and graduated responses to misuse as mechanisms for regulating membership, assigning responsibility, and responding to breaches. These formal arrangements clarified how access and accountability should work, but did not fully explain why members accepted trustees' judgement or continued to return and contribute over time. A long-standing participant (P1) used the emic term "*mana*" to describe "the inevitable quality of a charismatic person who is trusted as a leader of their community without being a bully, without having an inherited position of power." Participants linked the quality of *mana* (while not using the exact word) to reputational authority, sustained participation, reliability, and accountable contribution. Trustee elections could ratify this authority, but did not create it. However, other participants resisted the notion of leadership altogether. P9 stated that "trustees are not leaders," while P2 said that "there's no leaders, we're all members." This combination of resistance toward nominal leadership and appreciation for *mana* suggests that socially recognised authority operated beneath formal governance while remaining distinct from formal leadership.

Trusteeship also carried little material reward. P1 described trustees as people who "don't get power, you don't get any rewards," making the role "a very thankless task." Read alongside accounts of *mana*, the lack of material reward suggests the importance of recognition and commitment to the community in sustaining unpaid governance. The same reputational authority extended into conflict resolution. Members treated trustees as the default point of contact ("email the trustees," P2), ordinary misuse was usually handled through self-regulation, and repeated violations could lead to a ban. Formal structure provided a skeleton, but reputational authority made coordination credible and kept the skeleton from becoming empty procedure.

Sustained participation through mutual witnessing

Members often worked on separate projects rather than on one shared task. P4 described his robotics work in these terms: "I mainly work myself, like with the robots… they do their own robots, their robots… we don't exchange…but yeah, we share our ideas." The makerspace therefore resembled an open studio more than a structured project team. Individual work remained socially meaningful because it could be shown, noticed, discussed, and used as a reason to ask for or offer help. We describe this pattern as mutual witnessing, meaning that members make individually pursued work visible to one another, sustaining social connection without necessarily sharing a single project. Choosing what to make expressed autonomy, while developing and sharing an artefact offered opportunities to demonstrate competence and connect with others. Wikis, Discord conversations, induction records, and informal explanations extended this visibility by turning private experience into a circulating resource.

These accounts complement Han et al.'s (2017) model by showing how autonomy, competence, and relatedness were intertwined in descriptions of choosing, developing, and sharing projects. This is consistent with a constructionist account in which learning and identity are produced through the visible construction of shareable artefacts (Holbert et al., 2020). However, newcomers had uneven access to these opportunities. P9 recalled finding the space "a little bit cliquey…for a totally new [member]" and spending his first year "on the Discord just chatting to people" before feeling able to attend in person. Even after this online bridge, he felt that attending required a practical reason: "I can only go if I've got

something to do, if I've got a project I'm working on." P5, who toured but did not join, cited cost and distance. Together, these accounts distinguish the practical and social barriers to joining from the experiences associated with continued participation.

Selective absorption of AI

Participants did not position AI as either a saviour or a threat. They accepted bounded assistance with information search, routine logistics, coding, data processing, and early prototyping. P4 described allowing AI to produce a quick prototype and then revising it "to make it exactly what I want," while cautioning that "you need strong background knowledge to correct AI's mistakes." P2 similarly located responsibility with the user, who must check the information and remain "the human asking the right question." P8 summarised the stance as "AI is exciting, but I'm not excited about everything to do with AI." Across these accounts, retaining human judgement was, perhaps unsurprisingly, an important condition of acceptable AI use.

The sharpest formulation was P1's "paradox of AI." When an assistant offered to summarise an important email, he argued that "if an email is important enough to require a summary, it is important enough for me to read it thoroughly and comprehend it myself." The objection was not simply possible inaccuracy. His concern suggests that delegating comprehension to AI may bypass the attention involved in understanding shared information and taking responsibility for it. When discussing possible uses, participants envisaged AI in peripheral support roles, such as a help desk or a procedural information screen. These accounts emphasised bounded assistance rather than delegating induction or trustee decision-making to AI. P5, the visitor, linked reliance to respect: "if you start relying on the machine, everyone will not respect each other because respect comes from the expertise in this kind of community." AI use was treated as acceptable when its function was limited, transparent, and reversible, and as questionable when it substituted for traceable human judgement.

Discussion and implications

The findings answer the research questions by showing that, in participants' accounts, the makerspace's long-term continuity depends on the interaction between reputational authority, visible contribution, and negotiated participation, while AI was accepted for bounded support but limited when it displaced human comprehension, verification, or accountability. For the first research question, theories of commons governance and SDT can be read as interdependent rather than separate explanations. Commons governance helps explain how rules, trustees, and sanctions structured access and responsibility, while SDT helps explain how autonomy, competence, and relatedness were sustained through visible making, recognition, and belonging. The connection lies in visible contribution. Making and sharing work gave members opportunities to exercise autonomy, demonstrate competence, and gain recognition, while also giving others grounds to trust their judgement in the governance of shared resources. This interpretation complements makerspace continuance research (Han et al., 2017) by suggesting that visible making connects self-directed activity with recognition and belonging. It is grounded in nine participants at one site, so whether the same pattern appears elsewhere requires further study.

For the second research question, the study proposes extending the notion of commons from material resources toward epistemic and reputational resources. Drawing on the knowledge commons works of Hess and Ostrom (2007) and learning through participation described by Wenger (1998), the findings suggest that AI assistance becomes consequential when it replaces members' engagement with shared information. Participants expressed resistance to using AI when it might put shared meaning in danger (such as P1's objection to using AI to summarise an important email) and placed emphasis on checking AI outputs, which when read along with previous research showing that the introduction of AI shifts critical thinking toward information verification and oversight (e.g., Lee et al., 2025), identifies a more specific boundary for acceptable AI use within member-governed collaborative communities: AI may assist with information, but members remain responsible for comprehending, verifying, and acting on consequential material. Accountable judgement links both research questions: reputational authority rested on it, and participants were reluctant to delegate it to AI. Our theoretical

contribution is to bring attention and accountability to the centre of thinking around the governance of knowledge commons. The question is not only who can access shared information but also who undertakes the interpretive work needed to use it responsibly.

As AI adoption becomes part of everyday makerspace community practice, questions of governance increasingly concern who may use a tool, what evidence should be checked, whether its assumptions and sources can be scrutinised, and which decisions must remain attributable to individual members. Community guidance on these questions could preserve experimentation while protecting the judgement through which trust is built and competence is developed, and it is best treated as an extension of existing trust and accountability structures rather than as a separate technical policy. Questions of access are similarly shaped by how boundaries are drawn, since a formally open commons may still be difficult to enter when its norms remain tacit or when competence is recognised only after a newcomer has learned how to display it, as P9's account suggests. Legitimate peripheral participation (Lave & Wenger, 1991) may thus depend on having a reason to be present, consistent with evidence of informal exclusion in makerspace (Hedditch & Vyas, 2023).

Conclusion

In this member-governed makerspace, participants associated continuity with formal arrangements, reputational authority, and mutual witnessing of individual projects. Their accounts linked acceptable AI use to bounded assistance that preserved human judgement. For learning sciences, the case suggests that sustaining collaborative and technology supported learning within member-governed communities involves attending to the everyday practices through which work becomes visible, competence is recognised, and responsibility is negotiated.

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